Summary

Reproduction is a process by which the living beings propagate or duplicate their own kind. Reproduction can be broadly classified into asexual reproduction and sexual reproduction. In asexual reproduction fusion of gametes are not involved, but in sexual reproduction the formation and fusion of gametes occur. Different modes of asexual reproduction are fission, budding, fragmentation and regeneration. Fission is further divided into binary fission, multiple fission, sporulation and strobilation. According to the plane of fission different kinds of binary fission have been identified in different organisms. They are simple irregular binary fission, transverse binary fission, longitudinal binary fission and oblique binary fission. Multiple fission is the division of the parent into many small daughter cells simultaneously. Budding is another mode of asexual reproduction. The parent body produces one or more buds; each bud grows into a young one and may separate from the parent to lead a normal life. When many buds are formed on the outer surface of the parent, it is known as exogenous budding. Hundreds of buds are formed inside the cytoplasm and remain within the body of the parent, this process is called endogenous budding. Fragmentation is another mode of asexual reproduction. In fragmentation the body of the parent breaks into fragments (pieces). Each fragment has the potential to develop into a new individual. Regeneration is the development of the whole body of an organism from a small fragment. It is of two types namely restorative regeneration and reparative regeneration.

Various modes of sexual reproduction is seen in animals. In syngamy the fusion of two haploid gametes takes place to produce a zygote. The following kinds of syngamy is prevalent among the living organism. They are autogamy, exogamy, hologamy, paedogamy, merogamy, isogamy, anisogamy and conjugation. Parthenogenesis is the special type of sexual reproduction seen in animals. It is of two main types namely natural parthenogenesis and artificial parthenogenesis.

Evaluation

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1. In which type of parthenogenesis are only males produced?
   a) Arrhenotoky
   b) Thelytoky
   c) Amphitoky d) Both a and b

2. The mode of sexual reproduction in bacteria is by
   a) Formation of gametes
   b) Endospore formation
   c) Conjugation
   d) Zoospore formation

3. In which mode of reproduction variations are seen
   a) Asexual b) Parthenogenesis
   c) Sexual d) Both a and b

4. Assertion and reasoning questions:
   In each of the following questions there are two statements. One is assertion (A) and other is reasoning (R). Mark the correct answer as
   a. If both A and R are true and R is correct explanation for A
   b If both A and R are true but R is not the correct explanation for A
   c. If A is true but R is false
   d. If both A and R are false.

   I. Assertion: In bee society, all the members are diploid except drones.
   Reason: Drones are produced by parthenogenesis.
   a b c d

XII Std Biology-Zoology Chapter-1 EM.indd 8 23/12/2022 10:57:06